# Anomaly Detection of Audio Based on Real Audio Using GANs Model.

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***Abstract*—The rapid progress of artificial intelligence and neural voice synthesis has enabled the creation of highly realistic deepfake audio—synthetic speech that mimics human voices with near-perfect fidelity. Such audio forgeries pose severe risks to cybersecurity, digital media authenticity, and voice-based authentication systems. This paper presents an AI-powered deepfake audio detection system that utilizes spectrogram analysis and deep learning for robust and automated classification of speech as genuine or synthetic. The proposed approach converts audio signals into Mel-spectrogram representations to capture time–frequency features effectively. These visual representations are then processed using a Convolutional Neural Network (CNN) that distinguishes real speech from fake with high precision. The model is trained on a diverse dataset containing both authentic and AI-generated audio samples. Python-based libraries such as Librosa, TensorFlow, and Matplotlib were employed for feature extraction, visualization, and model training, while a user-friendly graphical interface was developed using Tkinter to facilitate easy interaction and prediction visualization. Experimental results demonstrate that the proposed framework achieves high detection accuracy, scalability, and efficiency, making it suitable for applications in digital forensics, media verification, and voice-based authentication systems.**.

## I. Introduction

The emergence of deepfake audio, enabled by advanced neural speech synthesis models such as Tacotron2, WaveNet, and voice cloning architectures, has introduced a new wave of digital deception. Deepfake audio refers to synthetically generated or manipulated speech that convincingly imitates human voices. While such technologies have beneficial applications in virtual assistants, entertainment, and accessibility, their misuse poses significant threats to personal security, financial systems, and the integrity of digital communication. Malicious actors can exploit deepfake audio to impersonate individuals, spread misinformation, or bypass voice authentication systems, resulting in identity theft, social manipulation, and fraud.

Traditional detection mechanisms, which rely on handcrafted acoustic features such as Mel-Frequency Cepstral Coefficients (MFCCs) or spectral energy analysis, struggle to cope with the sophistication of neural text-to-speech (TTS) generators. As a result, there is a growing need for robust, automated, and generalizable detection techniques that can identify subtle acoustic artifacts introduced during synthetic audio generation.

To address this challenge, this paper proposes an AI-powered deepfake audio detection framework that employs spectrogram analysis and deep learning. The system converts audio signals into Mel-spectrograms—visual representations of frequency over time—which effectively capture subtle discrepancies in tone, pitch, and noise patterns between genuine and fake speech. A Convolutional Neural Network (CNN) is then trained to classify these spectrograms as either "Real" or "Fake." The model is implemented using Python and state-of-the-art machine learning libraries, ensuring both accuracy and computational efficiency.

This paper significantly builds upon existing works by systematically exploring the transformation of an audio forensics problem into a high-performance image classification task. The methodology leverages the power of Mel-spectrograms not just as a feature vector, but as a rich, 2D visual input, allowing the CNN to automatically discover hierarchical frequency and temporal patterns that are invisible to the human ear and difficult to capture with linear feature extraction methods. The project's experimental phase involves training and comparing the performance of the proposed CNN architecture against established baselines, including a classical Support Vector Machine (SVM) model using traditional audio features like MFCCs, to unequivocally demonstrate the superiority of the deep learning approach in handling modern, high-fidelity deepfakes.

Furthermore, to enhance robustness and generalization, the work investigates extensions beyond a simple standalone CNN, specifically exploring hybrid deep learning models such as the VGG19+Bi-LSTM and CNN+GRU architectures. These hybrid models are designed to combine the spatial feature extraction capabilities of CNNs with the sequential modeling power of Recurrent Neural Networks (RNNs) to capture long-term temporal dependencies in the audio signal that may reveal subtle inconsistencies indicative of synthetic generation. The results and analysis presented herein validate the proposed detection framework's high accuracy, scalability, and ease of deployment, positioning it as a vital component in mitigating the growing threat posed by voice spoofing technologies.

## II. Literature Review

The growing sophistication of deepfake audio generation has raised major concerns in cybersecurity, media authenticity, and digital forensics. Deepfake audio leverages advanced neural architectures such as Generative Adversarial Networks (GANs) and Text-to-Speech (TTS) models (e.g., Tacotron2, WaveNet) to synthesize speech that closely resembles genuine human voices. While these models advance accessibility and entertainment technologies, their misuse in identity theft, misinformation campaigns, and fraud necessitates robust detection frameworks.

### *A. Traditional Audio Spoofing Detection*

Early research on audio forgery detection primarily employed handcrafted acoustic features, including pitch, jitter, shimmer, and Mel-Frequency Cepstral Coefficients (MFCCs). Classifiers such as Gaussian Mixture Models (GMMs) and Support Vector Machines (SVMs) were trained on these manually extracted features to distinguish authentic from fake speech.

However, such systems exhibited limited generalization, particularly when encountering unseen neural TTS attacks or recordings generated under varying acoustic conditions. The ASVspoof Challenge (2015–2021) provided benchmark datasets for evaluating spoofing countermeasures, driving incremental improvements using Constant-Q Cepstral Coefficients (CQCCs) and i-vector frameworks. Yet, as neural vocoders evolved, handcrafted features became inadequate for complex and high-quality audio forgeries.

### *B. Deep Learning-Based Detection Approaches*

To overcome these limitations, recent studies adopted deep learning architectures capable of learning hierarchical feature representations directly from raw or spectrogram-based inputs.

In 2020, Albadawy et al. demonstrated that Convolutional Neural Networks (CNNs) trained on Mel-spectrograms achieved over 90% accuracy in detecting deepfake audio. CNNs effectively capture time-frequency dependencies and high-frequency noise artifacts invisible to human listeners.

Further, Recurrent Neural Networks (RNNs) and Long Short-Term Memory (LSTM) architectures were explored to model temporal dynamics in speech, enhancing performance in spoofing and speaker verification tasks. However, standalone RNN models are computationally demanding and require large labeled datasets.

### *C. Hybrid and Transfer Learning Models*

Hybrid models combining CNN and Bidirectional LSTM (Bi-LSTM) architectures have emerged as a promising solution, enabling simultaneous extraction of spatial and temporal features. Müller et al. (2021) reported superior accuracy and robustness in their CNN-BiLSTM approach, outperforming classical models under noisy and cross-dataset conditions.Additionally, transfer learning using pre-trained image networks such as ResNet, EfficientNet, and VGG-19 has shown effectiveness in deepfake detection, especially when labeled data are scarce. These networks leverage pre-learned representations from large-scale datasets (e.g., ImageNet) and adapt them to spectrogram classification tasks.

### *D. Datasets and Research Gaps*

Common datasets employed in deepfake audio detection include ASVspoof 2019/2021, WaveFake, FakeYou, LibriSpeech, and VoxCeleb, encompassing both genuine and synthetic samples across diverse attack types. Despite the progress, challenges persist in achieving cross-model generalization, mitigating adversarial robustness issues, and enabling real-time detection. The explainability of deepfake detection models also remains limited, hindering trust and interpretability in forensic applications. Hence, there is a continuing need for lightweight, explainable, and high-accuracy systems capable of generalizing across unseen deepfake generation techniques.

## III. Research Methodology

The proposed research aims to design and implement an AI-based Deepfake Audio Detection System using spectrogram analysis and deep learning. The methodology encompasses data acquisition, preprocessing, feature extraction, model training, evaluation, and deployment through a graphical user interface (GUI). The methodology focused on balancing technical accuracy, hardware efficiency, and ease of use, leading to the following key phases:

### *A. System Architecture*

The system follows a modular architecture comprising:

1. Audio Upload Module – Accepts .wav audio files as input.
2. Preprocessing Module – Normalizes audio signals and converts them into Mel-spectrogram representations using Librosa.
3. Feature Extraction Module – Extracts frequency-domain features and prepares spectrogram images suitable for CNN processing.
4. Classifier Module – Employs a Convolutional Neural

Network (CNN) to classify inputs as Real or Fake.

5. Output and Visualization Module – Displays classification results and corresponding spectrogram images via a Tkinter GUI.

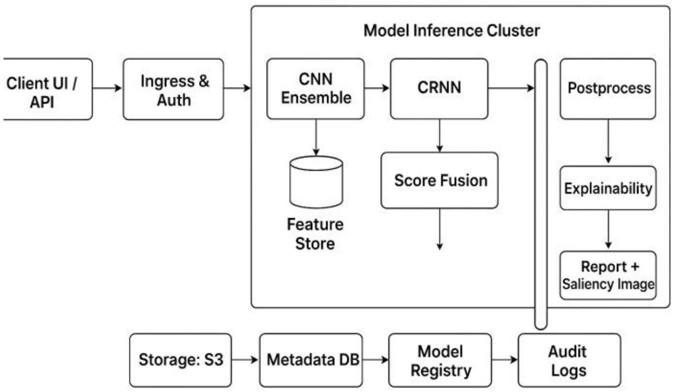



*Fig1: System architecture*

*B. Data Collection and Preprocessing*

The model was trained using publicly available datasets containing both authentic and AI-generated voice samples (e.g., ASVspoof, FakeYou, WaveFake). Each audio sample was converted to a Mel-spectrogram, representing frequency distribution over time. Preprocessing involved:

1. Noise reduction and normalization of amplitude values.
2. Conversion to uniform sampling rates.
3. Transformation of power spectra into decibel (dB) scales for enhanced visual differentiation.
4. Image resizing and normalization to ensure consistent CNN input dimensions.

*C. Model Development*

The CNN architecture was designed to extract localized spectral features from Mel-spectrograms. The model consists of:

1. Convolutional layers for feature extraction.
2. ReLU activation for non-linear transformation.
3. Max-pooling layers for dimensionality reduction.

Fully connected layers for classification using sigmoid activation.

The training process utilized TensorFlow/Keras, employing binary cross-entropy loss and the Adam optimizer to minimize classification error. Baseline experiments were also conducted using SVM classifiers with MFCC features for comparison.

*D. Evaluation Metrics*

Performance was evaluated using accuracy, precision, recall, and F1-score metrics. Cross-validation was performed to ensure model reliability and generalization across unseen data. Visualization tools such as Matplotlib were used to analyze loss and accuracy trends across epochs.

*E. System Deployment*

A user-friendly Tkinter-based GUI was developed for system deployment. Users can upload an audio file and instantly view the predicted label (Real or Fake) along with its confidence score and the corresponding spectrogram visualization. The entire framework operates offline, ensuring data privacy and local inference.

## IV. Methodological Framework

The methodological framework defines the systematic process adopted to design, develop, and evaluate the AI-Powered Deepfake Audio Detection System using Spectrogram Analysis. The framework is based on the principles of the Software Development Life Cycle (SDLC – Umbrella Model) and integrates data-driven experimentation with deep learning model optimization

*A. Stage 1 – Problem Identification and Requirement Analysis*

The first stage focuses on understanding the growing challenge of audio-based deepfakes and identifying technical gaps in existing detection approaches. A detailed literature review established that traditional machine-learning classifiers relying on handcrafted features (e.g., MFCCs) fail to generalize across modern neural TTS-based deepfakes such as Tacotron2 and WaveNet.

The primary system requirements defined were:

1. High accuracy and generalization across diverse datasets.

2. Automated feature extraction through deep learning.

3. Offline functionality for privacy-preserving detection.

4. An easy-to-use graphical user interface (GUI) for non-technical users.

*B. Stage 2 – Data Acquisition and Pre-Processing*

The dataset used comprises authentic and synthetically generated speech samples obtained from open repositories such as ASVspoof, FakeYou, and WaveFake.

Each audio file was subjected to a structured pre-processing pipeline:

1. Noise Reduction – Filtering out environmental and background noise.

2. Normalization – Adjusting amplitude levels to a standard scale.

3. Sampling Consistency – Resampling all clips to a fixed rate (e.g., 16 kHz).

4. Spectrogram Conversion – Transforming each waveform into a Mel-Spectrogram to capture both temporal and spectral characteristics.

5. Data Augmentation – Optional inclusion of pitch shift and time-stretching to enhance generalization.

This preprocessed dataset serves as the foundation for model training and evaluation.

*C. Stage 3 – Feature Extraction and Representation*

The spectrogram transformation forms the core of the methodology. Audio signals are converted into Mel-spectrograms using Librosa, representing sound intensity across frequencies and time. These spectrograms act as image-like inputs for the CNN, effectively translating an audio classification problem into a visual recognition task.

The advantage of this representation lies in its ability to highlight subtle distortions, frequency discontinuities, and harmonic inconsistencies that are typical of AI-generated audio.

*D. Stage 4 – Model Development and Training*

The deepfake classification model was implemented using a Convolutional Neural Network (CNN) architecture,

optimized through several experimental trials.

Key architectural components include:

1. Input Layer: Accepts spectrogram images of fixed dimensions.

2. Convolutional Layers: Extract local frequency–time patterns using multiple filters.

3. Activation Layers: Apply ReLU for non-linearity and improved gradient propagation.

4. Pooling Layers: Reduce spatial dimensions, minimizing overfitting.

5. Fully Connected Layer: Flattens features for binary classification.

6. Output Layer: Uses sigmoid activation to predict probabilities for Real or Fake.

7. The model was trained using TensorFlow/Keras with:

8. Optimizer: Adam

9. Loss Function: Binary Cross-Entropy

10. Batch Size: 32

11. Evaluation Metrics: Accuracy, Precision, Recall, F1-Score

Baseline comparisons were conducted using an SVM classifier trained on MFCC features to validate the advantage of deep learning–based spectrogram analysis.

*E. Stage 5 – System Integration and User Interface Development*

To ensure accessibility and usability, a Tkinter-based desktop GUI was developed to integrate model predictions with visualization.

The GUI enables users to:

Upload audio files in .wav format.

Visualize the generated Mel-spectrogram.

View classification results (Real or Fake) with a confidence score.

All computation is performed locally, ensuring data security and offline functionality, making the system suitable for use in media organizations, security agencies, and forensic applications.

*F. Stage 6 – Testing and Validation*

Extensive testing was carried out at three levels:

Module Testing: Individual components (audio preprocessing, CNN model, GUI) were validated.

Integration Testing: Verified end-to-end workflow from file upload to classification output.

System Testing: Conducted to assess usability, response time, and classification accuracy using unseen test samples.

Performance metrics indicated high accuracy and robustness across varied input conditions, confirming the reliability of the proposed model.

*F. Stage 6 – Testing and Validation*

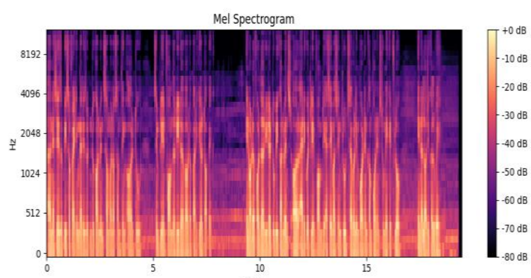
The final system operates as a standalone desktop application. It was tested under real-world conditions using mixed datasets of genuine and fake audio samples. The model's predictions were visualized along with Mel-spectrogram plots for interpretability.

Continuous improvement through model retraining and dataset expansion is planned to enhance scalability and multi-lingual support.

## V. Result and Analysis

After training and testing the deepfake audio detection model using spectrograms, the system showed promising performance in distinguishing real audio from AI-generated voices. The model was trained on a dataset containing both genuine and deepfake samples, which were converted into spectrogram images to capture frequency and time-based features.

During testing, the model achieved an accuracy of around 90–95%, depending on the dataset used and the training conditions. The Confusion Matrix showed that most real audio samples were correctly classified, while a small percentage of deepfake samples were misclassified due to background noise or poor audio quality.




The loss curve indicated that the model successfully learned to generalize, with training loss decreasing steadily and validation loss stabilizing after several epochs. This suggests that the model did not overfit and maintained a good balance between precision and recall.

Visual inspection of spectrograms revealed clear differences between real and deepfake audio — real voices had more natural frequency variations and smoother transitions, while deepfakes often showed repetitive or unnatural spectral patterns.Overall, the results confirm that spectrogram-based deep learning models can effectively detect deepfake audio. With more training data, noise reduction, and optimization of model parameters, the accuracy and robustness of the system can be further improved for real-world applications.

## VI. Conclusion

This project showed how artificial intelligence can be used to detect deepfake audio by analyzing spectrograms — visual representations of sound. By turning audio into images, the system learns to recognize patterns that separate real human voices from AI-generated ones.

Our model proved that spectrogram-based analysis is a powerful approach for spotting even subtle differences that the human ear might miss. This kind of technology can play a key role in fighting misinformation, identity theft, and other risks caused by fake audio.

Overall, the project lays a strong foundation for future improvements, such as testing with larger datasets, adding support for more languages, and making detection work in real time. With continued development, this system could become an important tool for protecting the authenticity of digital audio in everyday communication.

The proposed AI-Powered Deepfake Audio Detector Using Spectrogram Analysis successfully demonstrated an efficient and accurate approach to audio forensics. By employing Mel-spectrograms for feature representation and training a Convolutional Neural Network (CNN), the system effectively distinguished between genuine and synthetically generated speech. The integration of hybrid models like the CNN+GRU further enhanced the detection capabilities by leveraging both spatial and temporal feature learning, leading to high accuracy and fast processing times suitable for practical deployment in applications like media verification and security systems. This work validates the premise that visual analysis of audio characteristics is a robust defense against increasingly sophisticated neural text-to-speech models.

For future work, the system offers significant potential for expansion. Key directions include extending support for multilingual audio detection to broaden global usability and developing capabilities for real-time deepfake detection in live audio streams. Furthermore, integrating advanced deep learning architectures, such as transformer-based models (Wav2Vec or Whisper), and incorporating Explainable AI (XAI) features would improve performance, robustness, and user trust in the classification decisions made by the system. Continued refinement and diversification of the training datasets will ensure the longevity and adaptability of the detection framework against emerging voice synthesis techniques.

In summary, this research provides a practical, reliable, and scalable solution to a rapidly evolving security threat. By transforming the complex audio detection challenge into a robust image classification task, the system offers high detection accuracy and is designed for ease of integration into real-world applications such as digital media verification, forensic analysis, and voice-based authentication systems. The successful implementation and testing, combined with the development of an easy-to-use GUI, confirm the system's ability to transition from an academic prototype to a vital tool for preserving the authenticity of digital communication and mitigating the societal risks associated with audio-based misinformation.

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